E1

Why is frequency compensation omitted in comparator Integrated Circuits?

E2

Find the threshold voltages and sketch the voltage transfer characteristic for each of the Schmitt trigger circuits illustrated in Figure P12.8.
The output voltage levels of the comparators are +10 V and -10 V.

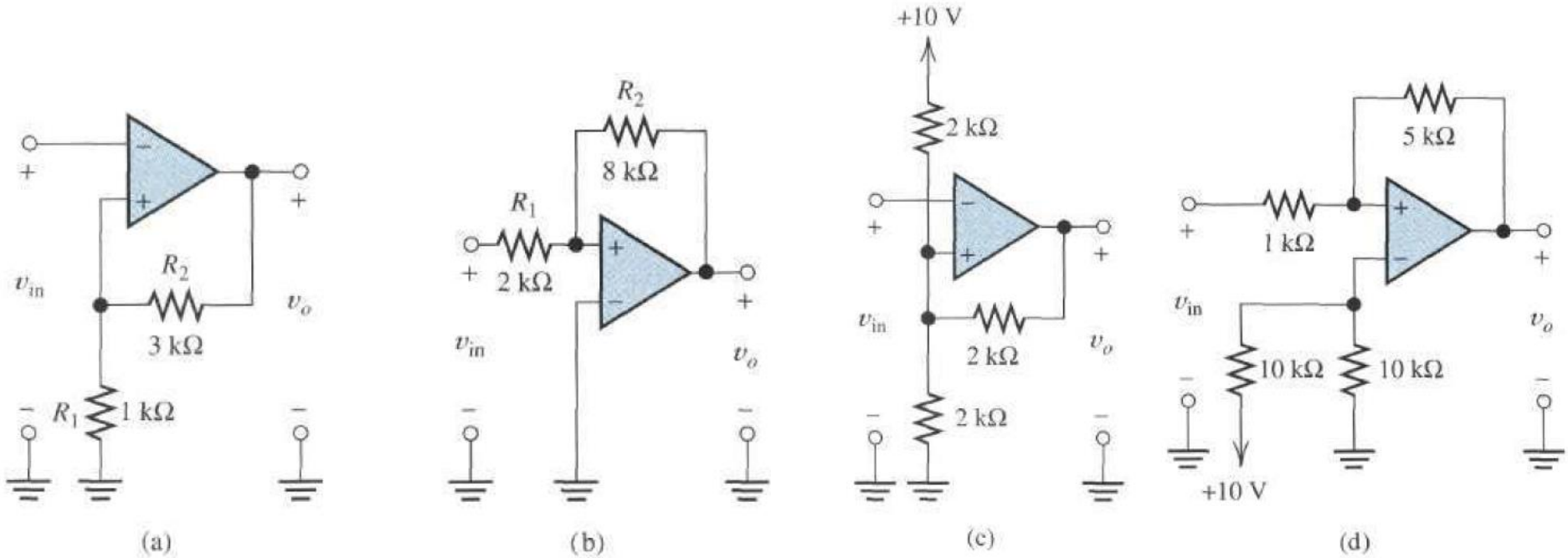


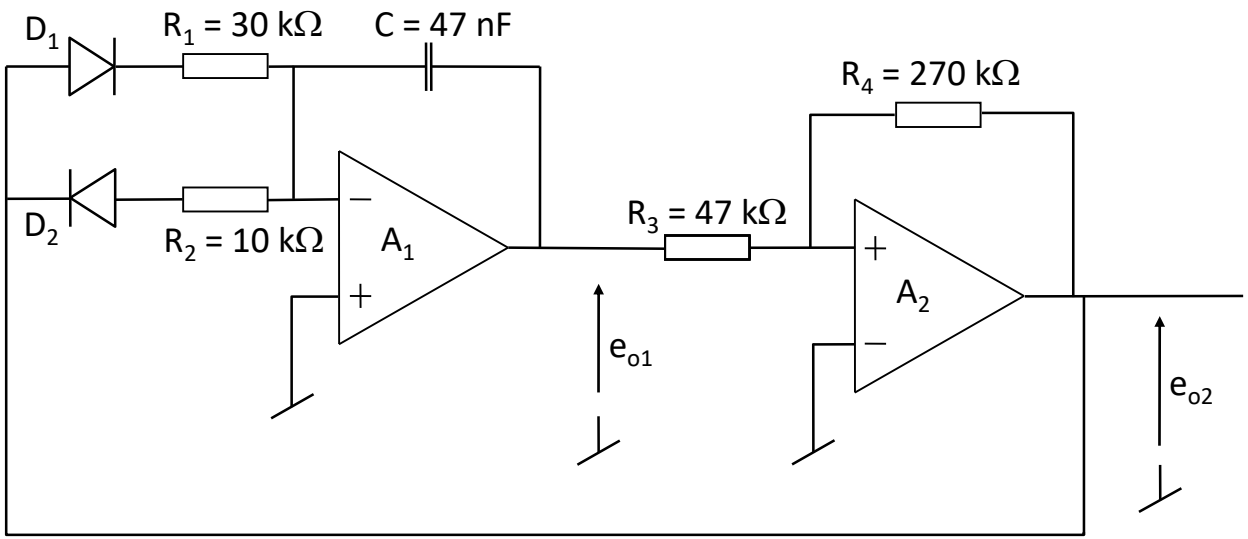
Figure P12.8

E3

Nedenstående diagram viser en integrator efterfulgt af en komparator. For A_1 og A_2 gælder følgende for mætningsspændingerne i samtlige spørgsmål:

$$V_{o\,SAT}^{+} = 14\ \text{V}; \qquad V_{o\,SAT}^{-} = -13\ \text{V}$$

Med hensyn til øvrige data kan A_1 og A_2 regnes ideelle. D_1 og D_2 kan regnes ideelle i spørgsmål 1 til og med spørgsmål 3.



(Opgave E3 fortsætter på næste side)

1) Bestem tærskelspændingerne (bogstavudtryk og talværdi) for komparatoren når e_{o1} kan betragtes som indgangssignal til denne.
   Skitsér e_{o2} som funktion af e_{o1} og angiv hysteresen V_H på skitsen samt et bogstavudtryk for denne.
2) Skitsér e_{o1} og e_{o2} som funktion af tiden og i indbyrdes sammenhæng (mindst to perioder). Niveauer og hældninger angives ved bogstavudtryk.
3) Beregn frekvensen af integratorens udgangssignal e_{o1}.

I spørgsmål 4 gælder kan der regnes med et spændingsfald i lederetningen for D_1 og D_2 på V_D = 0,7 V. Øvrige data er som tidligere angivet.

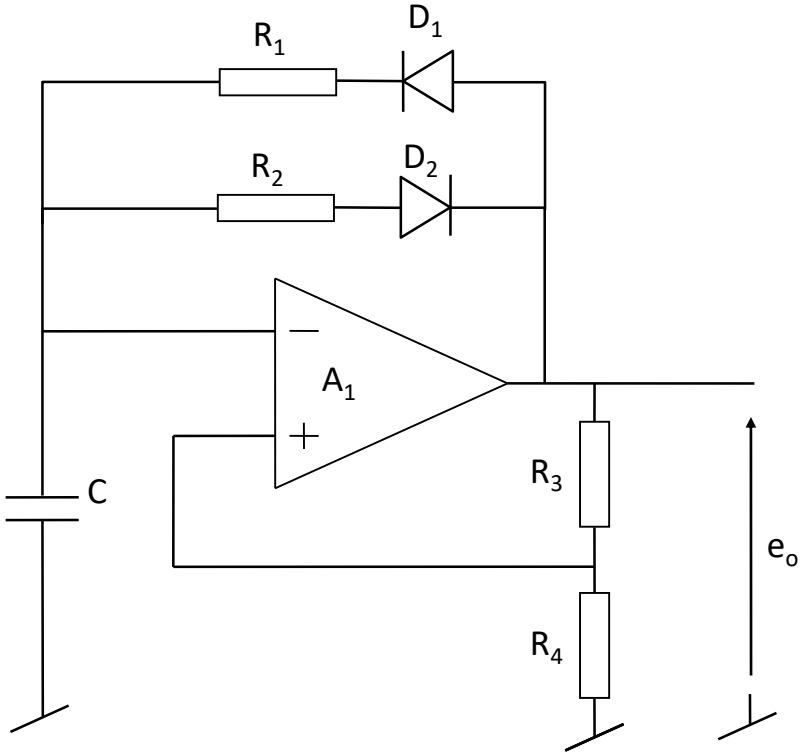
4) Hvilken procentisk fejl giver V_D anledning til på den i spørgsmål 3 beregnede frekvens?

E4

Figuren viser en astabil multivibrator med usymmetrisk udgangssignal. Multivibratoren ønskes dimensioneret således, at den afgiver et firkantsignal med frekvensen 1 kHz, og hvor puls/pause forholdet er 4:1. Ved "puls" forstås $V_o^+{}_{SAT}$ og ved "pause" $V_o^-{}_{SAT}$.

A_1 kan regnes ideel bortset fra: $V_o^+{}_{SAT} = -V_o^-{}_{SAT} = 10$ V

Bestem R_1 og R_2 og skitsér e_o, e^- og e^+ som funktion af tiden og i tidsmæssig sammenhæng (mindst to perioder). Niveauer angives ved bogstavudtryk.



$R_3 = 20$ kΩ

$R_4 = 5$ kΩ

$C = 0{,}01$ µF

D_1 og D_2 kan regnes ideelle

E5

Figuren viser en viser en kurveformgenerator (trekant/firkant). For A_1 og A_2 gælder:

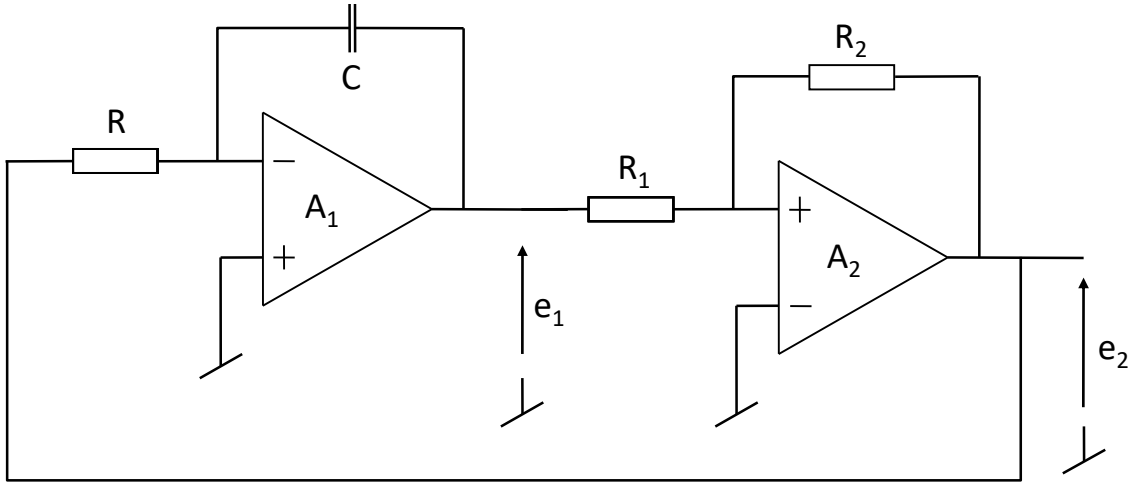
$$V_o^+{}_{SAT} = -V_o^-{}_{SAT}$$

Med hensyn til øvrige data kan de regnes ideelle i spørgsmål 1 og 2.

1) Skitsér e_1 og e_2 i sammenhæng som funktion af tiden og angiv samtlige hældningskoefficienter og spændingsniveauer (bogstavudtryk).

2) Bestem firkantsignalets frekvens (bogstavudtryk).

3) Vis med et bogstavudtryk hvorledes frekvensen bliver påvirket af offsetspænding V_{OS}, offsetstrøm I_{OS} og biasstrøm I_B for A_1. Bortset fra de nævnte fejlstørrelser har A_1 og A_2 samme data som i spørgsmål 1.

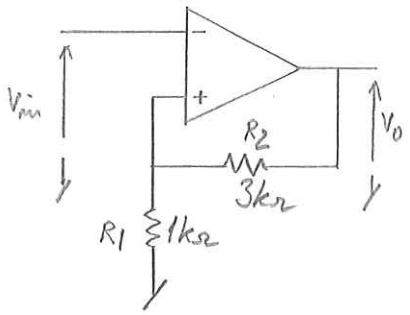


(E1) Frekvenskompensering er udeladt i komparator ICs, idet disse ikke skal anvendes med negativ feedback – dvs. i det lineære område.

(se s. 801)

(E2) For samtlige kredsløb gælder: $V_{0_{SAT}}^{+} = 10V$; $V_{0_{SAT}}^{-} = -10V$

(a)



$$e^- = V_{in}$$

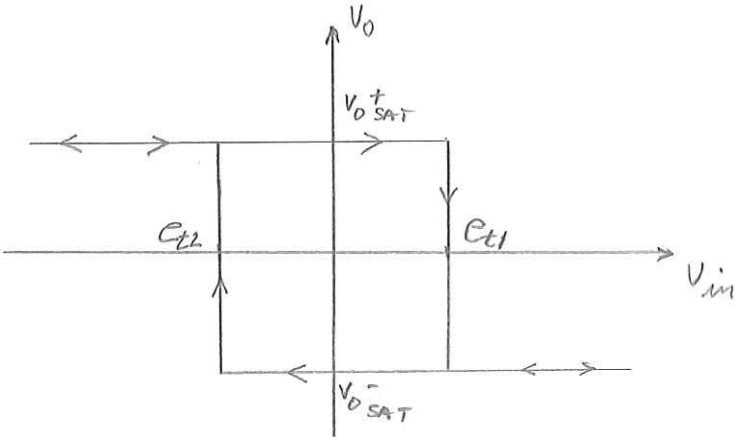
$$e^+ = V_0 \cdot \frac{R_1}{R_1+R_2}$$

$$\text{Komp.} \Rightarrow e^- = e^+ \Rightarrow$$

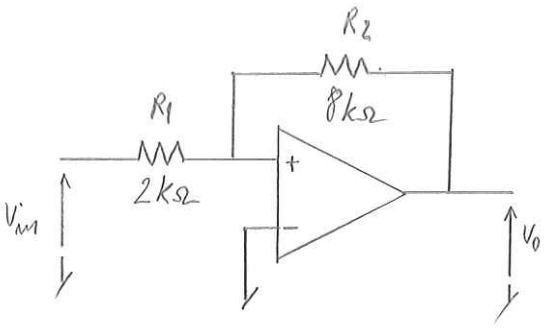
$$V_{in} = V_0 \cdot \frac{R_1}{R_1+R_2}$$

$$V_0 = V_{0_{SAT}}^{+} : \quad e_{t_1} = \frac{R_1}{R_1+R_2} \cdot V_{0_{SAT}}^{+} = \frac{1}{1+3} \cdot 10 = \underline{\underline{2{,}5V}}$$

$$V_0 = V_{0_{SAT}}^{-} : \quad e_{t_2} = \frac{R_1}{R_1+R_2} \cdot V_{0_{SAT}}^{-} = \frac{1}{1+3} \cdot (-10) = \underline{\underline{-2{,}5V}}$$



(b)



$e^- = 0V$

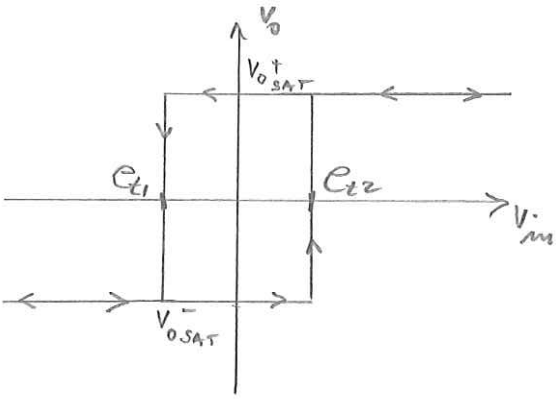
$e^+ = V_{in} \frac{R_2}{R_1+R_2} + V_0 \frac{R_1}{R_1+R_2}$

Komp. $\Rightarrow$ $e^- = e^+$ $\Rightarrow$

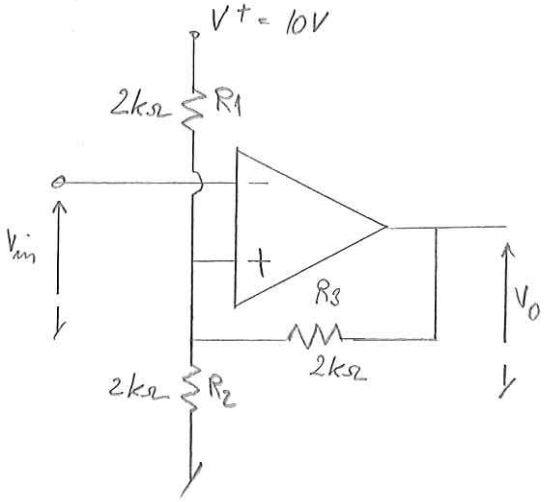
$V_{in} = -\frac{R_1}{R_2} \cdot V_0$

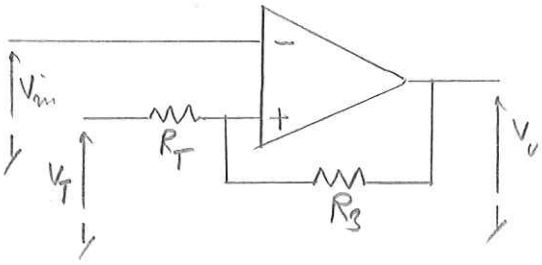
$V_0 = V_{0SAT}^+$: $e_{t_1} = -\frac{R_1}{R_2} \cdot V_{0SAT}^+ = -\frac{2}{8} \cdot 10 = \underline{\underline{-2{,}5V}}$

$V_0 = V_{0SAT}^-$: $e_{t_2} = -\frac{R_1}{R_2} \cdot V_{0SAT}^- = -\frac{2}{8} \cdot (-10) = \underline{\underline{2{,}5V}}$



(c)





$R_T = R_1 \parallel R_2 = 1k\Omega$

$V_T = \frac{R_2}{R_1+R_2} \cdot V^+ = 5V$

$$e^- = V_{in}$$

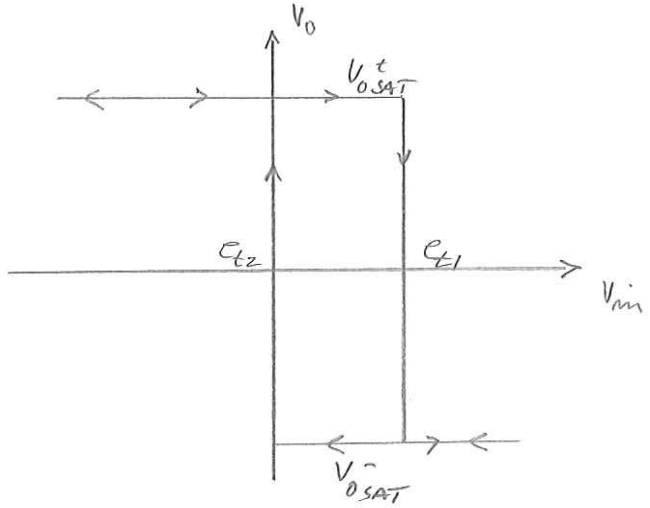
$$e^+ = V_T \frac{R_3}{R_T+R_3} + V_0 \frac{R_T}{R_T+R_3}$$

Komp. $\Rightarrow e^- = e^+ \Rightarrow$

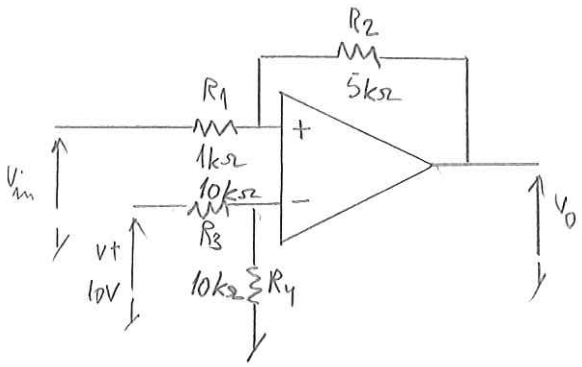
$$V_{in} = V_T \frac{R_3}{R_T+R_3} + V_0 \frac{R_T}{R_T+R_3}$$

$$V_0 = V_{0SAT}^+ : \quad e_{t1} = V_T \frac{R_3}{R_T+R_3} + V_{0SAT}^+ \frac{R_T}{R_T+R_3} = \underline{\underline{6{,}67\ V}}$$

$$V_0 = V_{0SAT}^- : \quad e_{t2} = V_T \frac{R_3}{R_T+R_3} + V_{0SAT}^- \frac{R_T}{R_T+R_3} = \underline{\underline{0\ V}}$$



(d)



$$e^- = V^+ \frac{R_4}{R_3+R_4}$$

$$e^+ = V_{in} \cdot \frac{R_2}{R_1+R_2} + V_0 \frac{R_1}{R_1+R_2}$$

Komp. $\Rightarrow e^+ = e^- \Rightarrow$

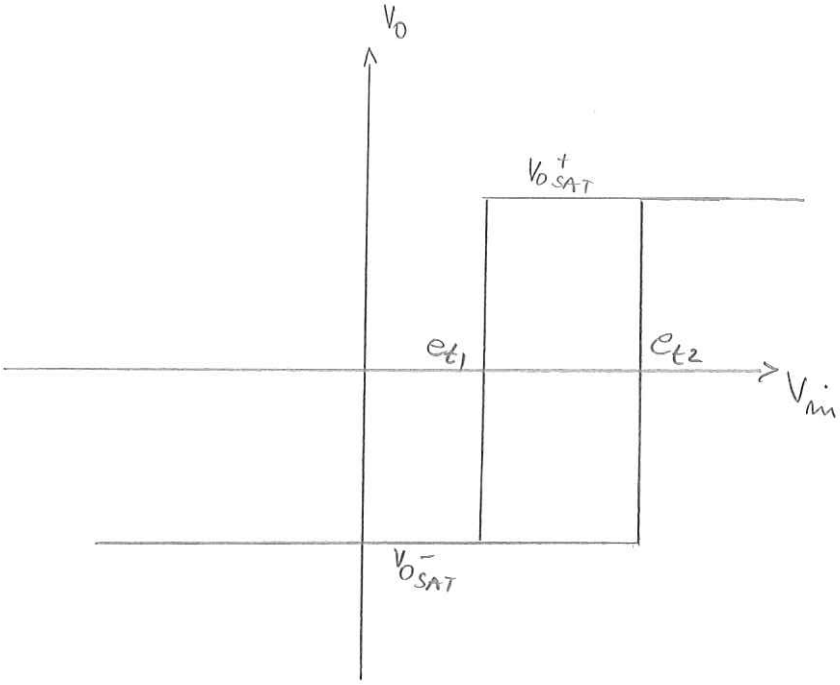
$$V^+ \frac{R_4}{R_3+R_4} = V_{in} \frac{R_2}{R_1+R_2} + V_0 \frac{R_1}{R_1+R_2}$$

$\Downarrow$

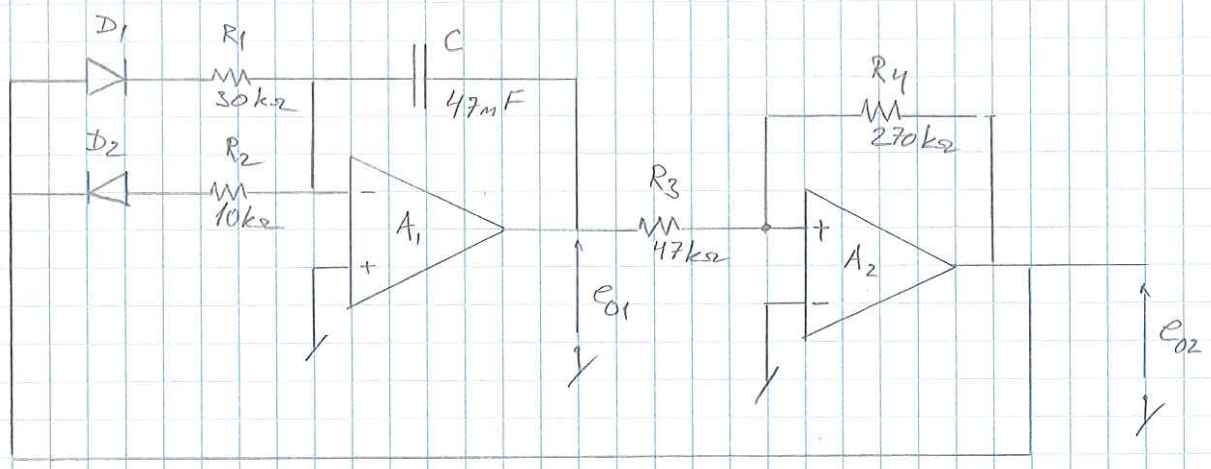
$$V_{in} = V^+ \frac{R_4}{R_3+R_4} \cdot \frac{R_1+R_2}{R_2} - V_0 \cdot \frac{R_1}{R_2}$$

$$V_0 = V_{0SAT}^+ : e_{t1} = V^+ \frac{R_4}{R_3+R_4} \frac{R_1+R_2}{R_2} - V_{0SAT}^+ \frac{R_1}{R_2} = 10 \cdot \frac{10}{10+10} \cdot \frac{1+5}{5} - 10 \cdot \frac{1}{5} = \underline{\underline{4\ V}}$$

$$V_0 = V_{0SAT}^- : e_{t2} = V^+ \frac{R_4}{R_3+R_4} \frac{R_1+R_2}{R_2} - V_{0SAT}^- \frac{R_1}{R_2} = 10 \cdot \frac{10}{10+10} \cdot \frac{1+5}{5} - (-10) \cdot \frac{1}{5} = \underline{\underline{8\ V}}$$


V_0
$V_{0\,SAT}^{+}$
e_{t1}
e_{t2}
V_{in}
$V_{0\,SAT}^{-}$

E3



A_1 og A_2 : $V^+_{0SAT} = 14\,V$; $V^-_{SAT} = -13\,V$ – ideelle mht. øvrige data

D_1 og D_2 : Ideelle i sp. 1-3 incl.

Sp. 1 A_2 : $e^+ = e_{01} \cdot \frac{R_4}{R_3+R_4} + e_{02} \frac{R_3}{R_3+R_4}$

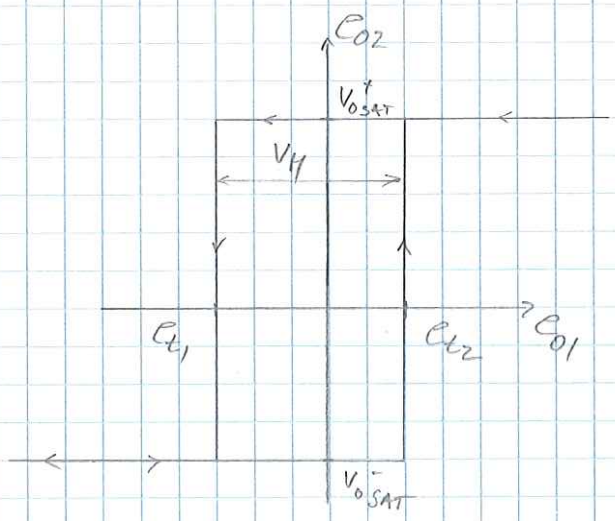
$e^- = 0$

Komp. $\Rightarrow e^- = e^+ \Rightarrow$

$$e_{01} \frac{R_4}{R_3+R_4} + e_{02} \frac{R_3}{R_3+R_4} = 0 \quad \Rightarrow \quad e_{01} = -\frac{R_3}{R_4} \cdot e_{02}$$

$e_{02} = V^+_{0SAT}$: $e_{t_1} = -\frac{R_3}{R_4} \cdot V^+_{0SAT} = -\frac{47}{270} \cdot 14 = \underline{\underline{-2{,}44\,V}}$

$e_{02} = V^-_{0SAT}$: $e_{t_2} = -\frac{R_3}{R_4} \cdot V^-_{0SAT} = -\frac{47}{270} \cdot (-13) = \underline{\underline{2{,}26\,V}}$

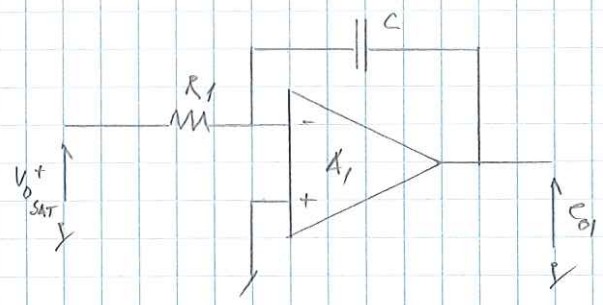


$$V_H = e_{t_2} - e_{t_1}$$

$$= -\frac{R_3}{R_4} \cdot V^-_{0SAT} - \left(-\frac{R_3}{R_4} \cdot V^+_{0SAT}\right)$$

$$= \underline{\underline{\frac{R_3}{R_4}\left(V^+_{0SAT} - V^-_{0SAT}\right)}}$$

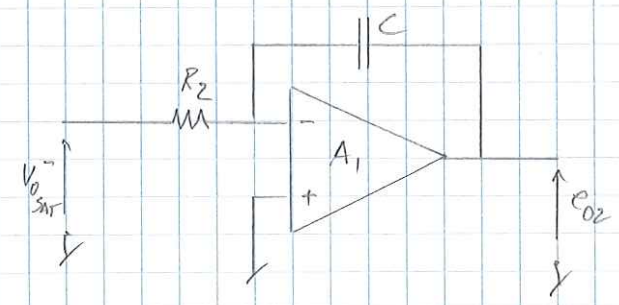
Sp. 2 $e_{o2} = V_{o\,SAT}^{+} \Rightarrow D_1$ "ON" ; D_2 "OFF" $\Rightarrow$



$$e_{o1} = -\frac{1}{CR_1} \cdot \int V_{o\,SAT}^{+}\, dt \Rightarrow$$

$$\frac{de_{o1}}{dt} = -\frac{V_{o\,SAT}^{+}}{CR_1} = \alpha_1$$

$e_{o2} = V_{o\,SAT}^{-} \Rightarrow D_1$ "OFF" ; D_2 "ON" $\Rightarrow$

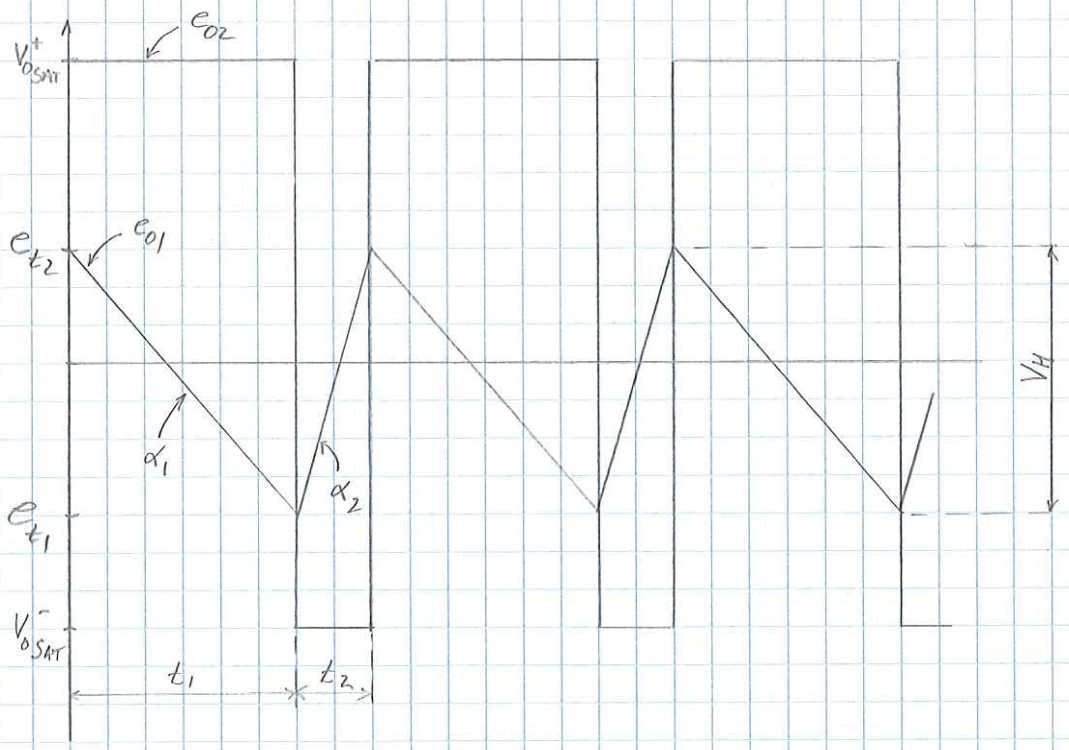


$$e_{o1} = -\frac{1}{CR_2} \int V_{o\,SAT}^{-}\, dt \Rightarrow$$

$$\frac{de_{o1}}{dt} = -\frac{V_{o\,SAT}^{-}}{CR_2} = \alpha_2$$

$$\alpha_1 = -\frac{V_{o\,SAT}^{+}}{CR_1} = -\frac{14}{47 \cdot 10^{-9} \cdot 30 \cdot 10^{3}} \cong -9929{,}078 \ V/s$$

$$\alpha_2 = -\frac{V_{o\,SAT}^{-}}{CR_2} = -\frac{(-13)}{47 \cdot 10^{-9} \cdot 10 \cdot 10^{3}} \cong 27660 \ V/s$$



Sp.3 Frekvensen af e_{01} : $f = \frac{1}{t_1 + t_2}$

$$t_1 = \frac{V_H}{|\alpha_1|} \quad ; \quad t_2 = \frac{V_H}{|\alpha_2|}$$

$$V_H = \frac{R_3}{R_4}\left(V_{0\,SAT}^{+} - V_{0\,SAT}^{-}\right) = \frac{47}{270}\cdot(14-(-13)) = 4{,}700\ V$$

$$t_1 = \frac{4{,}700}{9929{,}078} = 473{,}36\ \mu s$$

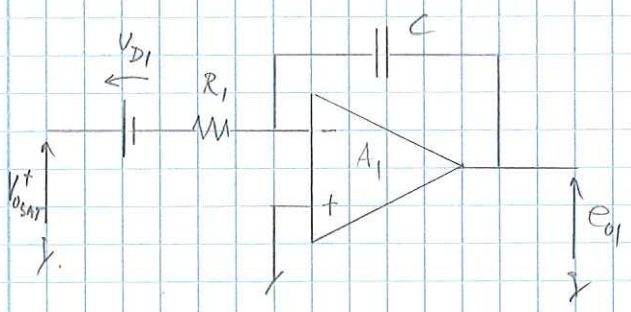
$$t_2 = \frac{4{,}700}{27660} = 169{,}92\ \mu s$$

$$f = \frac{1}{473{,}36\cdot 10^{-6} + 169{,}92\cdot 10^{-6}} = 1{,}5545\cdot 10^{3} \approx \underline{\underline{1{,}55\ kHz}}$$

Sp.4 I sp.4 gælder for D_1 og D_2 : $V_D = 0{,}7\ V$ (spændingsfald i ledetretning)

Nu er situationen som følger

$e_{02} = V_{0\,SAT}^{+} \Rightarrow D_1$ "ON", D_2 "OFF"



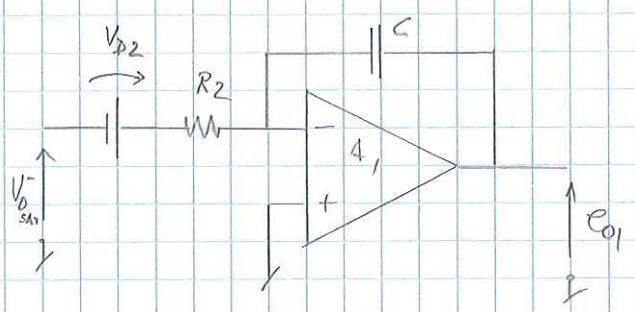
$$e_{01} = -\frac{1}{CR_1}\int\left(V_{0\,SAT}^{+} - V_{D1}\right)dt \Rightarrow$$

$$\frac{de_{01}}{dt} = -\frac{V_{0\,SAT}^{+} - V_{D1}}{CR_1} = \alpha_1'$$

$$\alpha_1' = -\frac{14 - 0{,}7}{47\cdot 10^{-9}\cdot 30\cdot 10^{3}}$$

$$= -9432{,}624\ V/s$$

$e_{02} = V_{0\,SAT}^{-} \Rightarrow D_1$ "OFF", D_2 "ON"



$$e_{01} = -\frac{1}{CR_2}\int\left(V_{0\,SAT}^{-} + V_{D2}\right)dt \Rightarrow$$

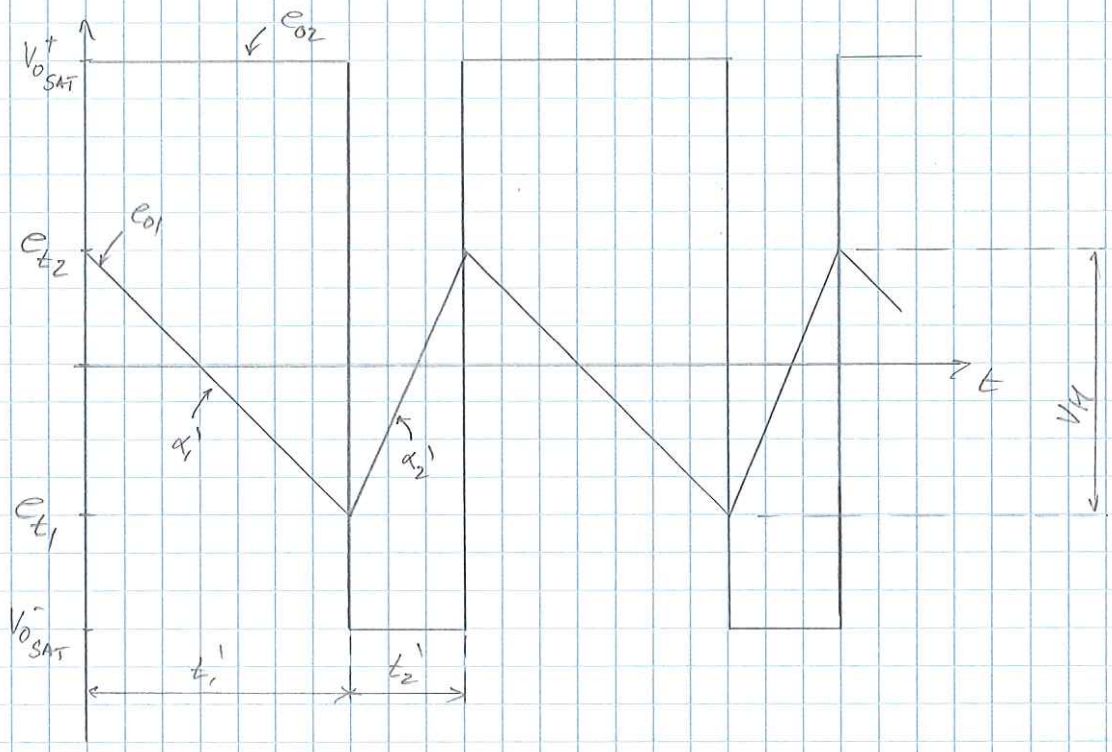
$$\frac{de_{01}}{dt} = -\frac{V_{0\,SAT}^{-} + V_{D2}}{CR_2} = \alpha_2'$$

$$\alpha_2' = -\frac{-13 + 0{,}7}{47\cdot 10^{-9}\cdot 10\cdot 10^{3}}$$

$$= 26170{,}21\ V/s$$

Tærskelspændingerne for komparatoren er uændrede, hvilket giver følgende skitse:

NB! $|\alpha_1| > |\alpha_1'|$; $|\alpha_2| > |\alpha_2'|$



$$f' = \frac{1}{t_1' + t_2'}$$

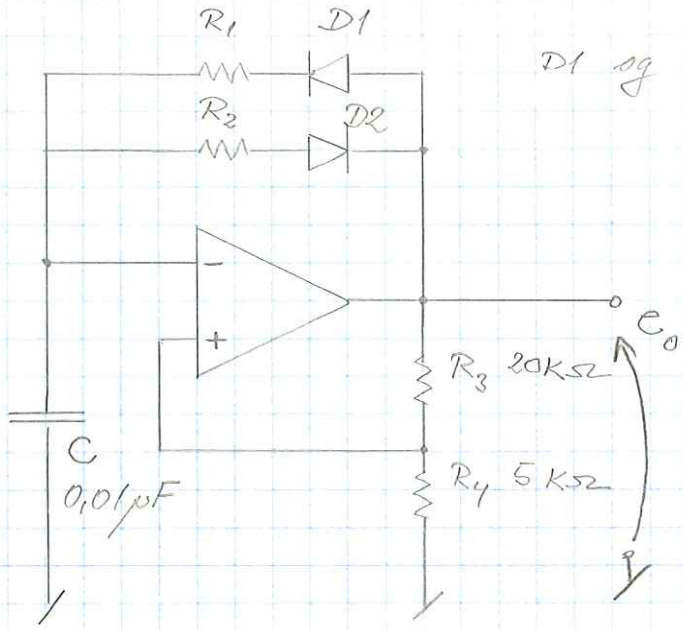
$$t_1' = \frac{V_H}{|\alpha_1'|} = \frac{4{,}700}{9432{,}624} = 498{,}27\ \mu s$$

$$t_2' = \frac{V_H}{|\alpha_2'|} = \frac{4{,}700}{26170{,}21} = 179{,}59\ \mu s$$

$$f' = \frac{1}{t_1' + t_2'} = \frac{1}{498{,}27 \cdot 10^{-6} + 179{,}59 \cdot 10^{-6}} = 1475{,}23\ Hz$$

Procentisk fejl

$$\frac{f' - f}{f} = \frac{1475{,}23 - 1554{,}5}{1554{,}5} = -5{,}099 \cdot 10^{-2} \approx \underline{\underline{-5{,}1\%}}$$

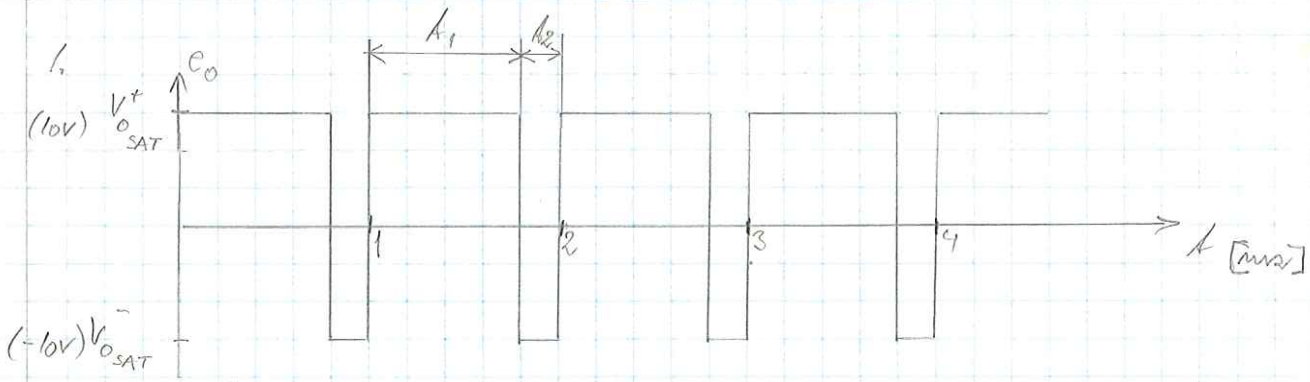


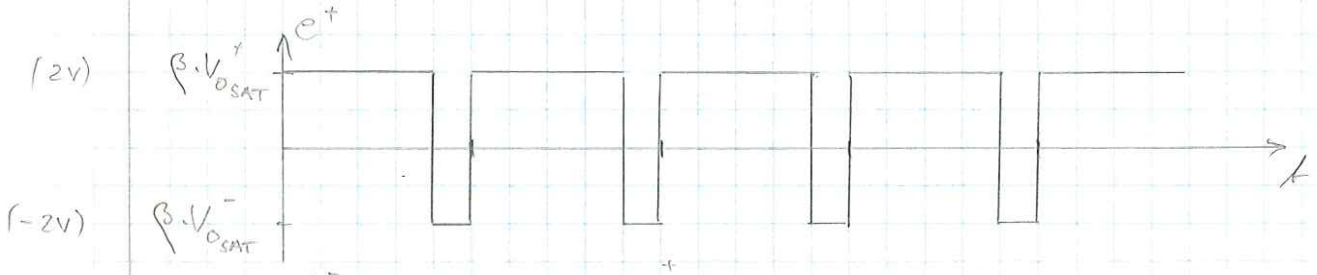
D1 og D2 er ideelle

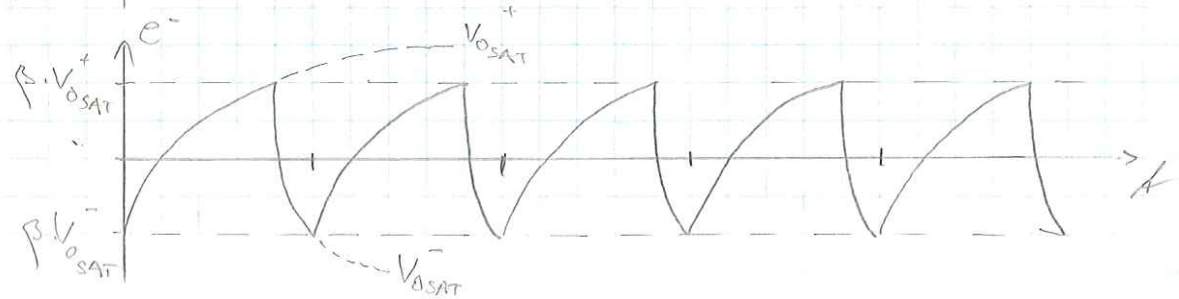
Op. amp. har metningsspændinger $V_{OSAT}^{+} = 10V$ og $V_{OSAT}^{-} = -10V$ (i øvrigt ideel)

Krav: $f = 1\,kHz \Rightarrow T = \frac{1}{f} = 10^{-3}s = 1ms$

pulz (V_{OSAT}^{+}) / pause $(V_{OSAT}^{-}) = 4/1$







2.

$$\beta = \frac{R_4}{R_3+R_4} = \frac{5}{20+5} = 0{,}2$$

$$t_1 = \frac{4}{5}\cdot T = \frac{4}{5}\cdot 10^{-3} = 0{,}8\,ms$$

$$t_2 = \frac{1}{5}\cdot T = \frac{1}{5}\cdot 10^{-3} = 0{,}2\,ms$$

I perioden t_1 oplades C (D1 „on" ; D2 „off")
I perioden t_2 aflades C (D1 „off" ; D2 „on")
Vi har da

$$t_1 = CR_1 \ln\frac{1+\beta}{1-\beta} \Leftrightarrow R_1 = t_1 \cdot \frac{1}{C\ln\frac{1+\beta}{1-\beta}}$$

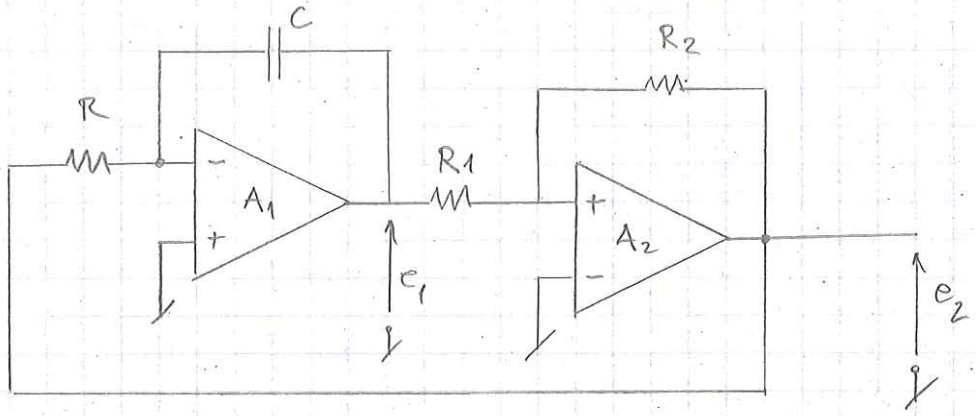
$$t_2 = CR_2 \ln\frac{1+\beta}{1-\beta} \Leftrightarrow R_2 = t_2 \cdot \frac{1}{C\ln\frac{1+\beta}{1-\beta}}$$

og ved indsættelse:

$$R_1 : 0{,}8\cdot 10^{-3}\,\frac{1}{0{,}01\cdot 10^{-6}\cdot \ln\frac{1+0{,}2}{1-0{,}2}} = \underline{\underline{197\,k\Omega}}$$

$$R_2 : 0{,}2\cdot 10^{-3}\cdot\frac{1}{0{,}01\cdot 10^{-6}\cdot \ln\frac{1+0{,}2}{1-0{,}2}} = \underline{\underline{49{,}3\,k\Omega}}$$

E5



Sp. 1 A_2 : $e^- = 0$

$$e^+ = e_1 \cdot \frac{R_2}{R_1+R_2} + e_2 \frac{R_1}{R_1+R_2}$$

Komp. $\Rightarrow e^+ = e^- \Rightarrow e_1 \frac{R_2}{R_1+R_2} + e_2 \frac{R_1}{R_1+R_2} = 0 \Rightarrow e_1 = -\frac{R_1}{R_2} \cdot e_2$

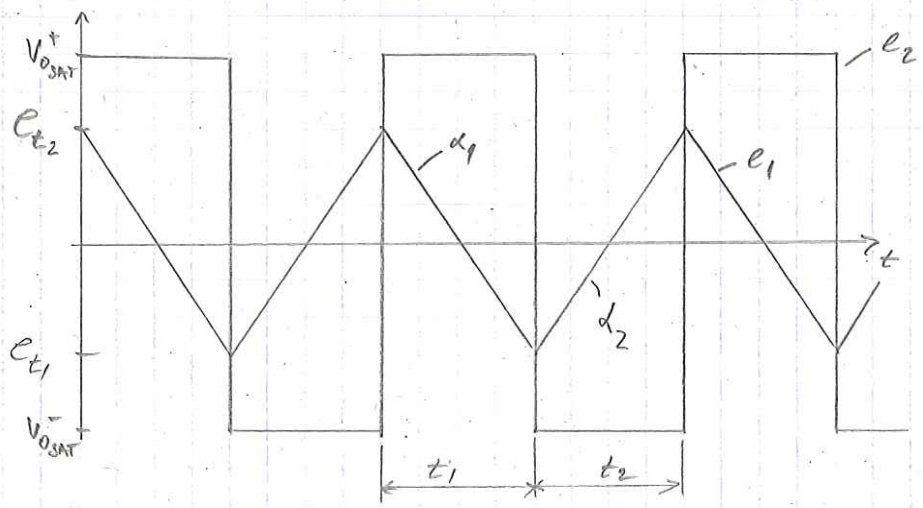
$e_2 = V_{0SAT}^+$: $e_{t_1} = -\frac{R_1}{R_2} \cdot V_{0SAT}^+$

$e_2 = V_{0SAT}^-$: $e_{t_2} = -\frac{R_1}{R_2} \cdot V_{0SAT}^-$

A_1 : $e_1 = -\frac{1}{CR} \int e_2 \, dt \Rightarrow \frac{de_1}{dt} = -\frac{e_2}{CR}$

$e_2 = V_{0SAT}^+$: $\frac{de_1}{dt} = -\frac{V_{0SAT}^+}{CR} = \alpha_1$

$e_2 = V_{0SAT}^-$: $\frac{de_1}{dt} = -\frac{V_{0SAT}^-}{CR} = \alpha_2$

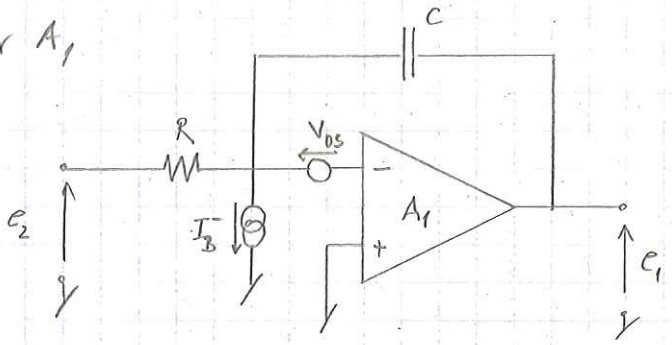


Sp. 2 Symmetri $\Rightarrow t_1 = t_2 = \frac{e_{t_2} - e_{t_1}}{|\alpha_1|} = \frac{-\frac{R_1}{R_2} \cdot V_{0SAT}^- - \left(-\frac{R_1}{R_2} \cdot V_{0SAT}^+\right)}{\frac{V_{0SAT}^+}{CR}} = \frac{2 \cdot CR \cdot R_1}{R_2}$

$(V_{0SAT}^+ = -V_{0SAT}^-)$

$$f = \frac{1}{t_1 + t_2} = \frac{1}{2 \cdot t_1} = \underline{\underline{\frac{R_2}{4 \cdot CR \cdot R_1}}}$$

Ekv. diagr. for A_1



$$I_B^- = I_B - \tfrac{1}{2} I_{OS}$$

$$V_{O\,SAT}^+ = -V_{O\,SAT}^-$$

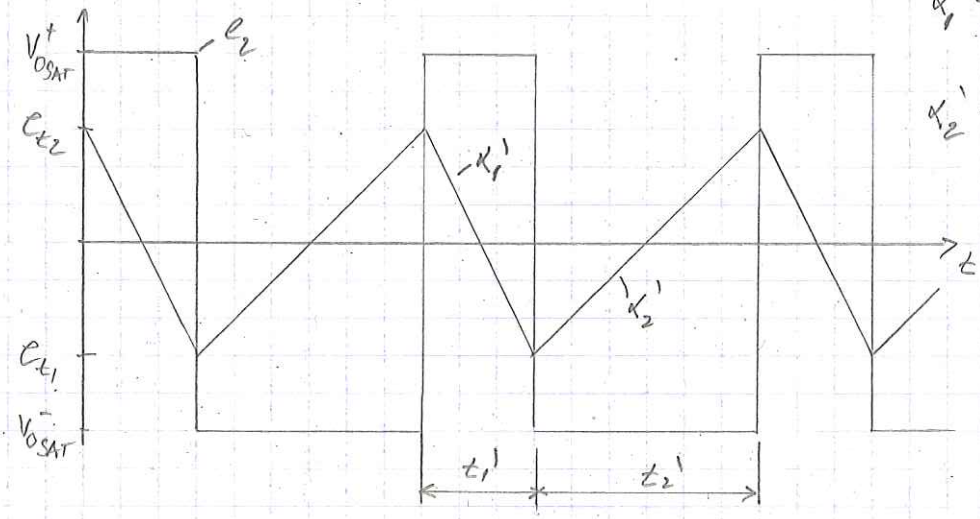
$$e_1 = -\frac{1}{CR}\int e_2\,dt + \frac{1}{CR}\int V_{OS}\,dt + \frac{1}{C}\int I_B^-\,dt + V_{OS} \quad \Rightarrow$$

$$\frac{de_1}{dt} = \underset{\substack{\uparrow \\ \text{ideelt}}}{-\frac{e_2}{CR}} + \underbrace{\frac{V_{OS}}{CR} + \frac{I_B^-}{C}}_{\text{fejl (- holdning / - drift)}}$$

Alt efter fejlene fortegn (dual) vil $\frac{de_1}{dt}$ henholdsvis forøges / formindskes i t_1 og t_2.

$$\alpha_1' = -\frac{V_{O\,SAT}^+}{CR} + \frac{V_{OS}}{CR} + \frac{I_B^-}{C}$$

$$\alpha_2' = -\frac{V_{O\,SAT}^-}{CR} + \frac{V_{OS}}{CR} + \frac{I_B^-}{C}$$



$$t_1' = \frac{e_{t_1} - e_{t_2}}{\alpha_1'} = \frac{-\frac{R_1}{R_2}\cdot V_{O\,SAT}^+ - \left(-\frac{R_1}{R_2}\cdot V_{O\,SAT}^-\right)}{-\frac{V_{O\,SAT}^+}{CR} + \frac{V_{OS}}{CR} + \frac{I_B^-}{C}} = \frac{\left(V_{O\,SAT}^+ - V_{O\,SAT}^-\right)\frac{R_1}{R_2}\cdot CR}{V_{O\,SAT}^+ - V_{OS} - I_B^-\cdot R} = \frac{2\cdot CR\cdot\frac{R_1}{R_2}}{1 - \frac{V_{OS} + I_B^-\cdot R}{V_{O\,SAT}^+}}$$

$$t_2' = \frac{e_{t_2} - e_{t_1}}{\alpha_2'} = \frac{-\frac{R_1}{R_2}\cdot V_{O\,SAT}^- - \left(-\frac{R_1}{R_2}\cdot V_{O\,SAT}^+\right)}{-\frac{V_{O\,SAT}^-}{CR} + \frac{V_{OS}}{CR} + \frac{I_B^-}{C}} = \frac{\left(V_{O\,SAT}^+ - V_{O\,SAT}^-\right)\frac{R_1}{R_2}\cdot CR}{V_{O\,SAT}^+ + V_{OS} + I_B^-\cdot R} = \frac{2\cdot CR\cdot\frac{R_1}{R_2}}{1 + \frac{V_{OS} + I_B^-\cdot R}{V_{O\,SAT}^+}}$$

$$x = \frac{V_{OS} + I_B^-\cdot R}{V_{O\,SAT}^+} \Rightarrow f' = \frac{1}{\frac{2CR\frac{R_1}{R_2}}{1-x} + \frac{2CR\frac{R_1}{R_2}}{1+x}} = \frac{(1-x)(1+x)}{2CR\frac{R_1}{R_2}\left(1+x+1-x\right)}$$

$$= \frac{1-x^2}{4CR\frac{R_1}{R_2}} = \frac{R_2}{4CRR_1} - \frac{R_2}{4CRR_1}\cdot x^2 \Rightarrow$$

$$f' < f_{ideel} = \frac{R_2}{4CRR_1} \quad \text{jf. sp. 2.}$$